

WAKEFIELD, VA, USA

WMO: 724019

Lat: 36.983N

Lon: 77.001W

Elev: 34

StdP: 100.92

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 93773

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.0	-5.7	-17.6	0.8	-3.9	-14.9	1.0	-2.0	7.4	12.2	6.7	10.1	1.0	350	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	34.5	24.6	33.2	24.2	32.1	23.8	26.1	32.0	25.5	31.1	25.0	30.1	2.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	19.5	28.5	23.9	18.9	28.0	23.4	18.3	27.5	81.0	32.1	78.3	31.3	75.9	30.2	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.2	5.3	4.7	-13.2	36.8	4.6	1.6	-16.6	38.0	-19.3	39.0	-21.9	39.9	-25.3	41.1
			-14.0	27.6	4.4	0.7	-17.1	28.1	-19.7	28.5	-22.1	28.9	-25.3	29.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	4.3	5.8	9.4	15.1	19.8	24.1	26.0	25.4	22.1	15.9	10.2	6.4
	DBStd	8.92	5.99	5.40	5.54	4.96	4.19	3.21	2.63	2.44	3.45	4.64	4.85	5.33
	HDD10.0	629	196	139	82	10	0	0	0	0	0	6	57	140
	HDD18.3	1957	436	350	280	122	36	2	0	0	9	103	248	371
	CDD10.0	2607	18	23	64	164	303	422	494	476	362	190	62	28
	CDD18.3	893	1	1	4	25	81	174	236	218	121	29	3	1
	CDH23.3	8209	1	4	43	260	716	1674	2366	1978	932	214	17	3
	CDH26.7	3232	0	0	4	68	232	694	1054	821	308	48	2	0
Wind	WSAvg	1.6	1.8	2.0	2.1	2.1	1.7	1.5	1.4	1.1	1.2	1.3	1.5	1.6
Precipitation	PrecAvg	1163	86	67	108	82	92	110	129	118	122	84	76	88
	PrecMax	1549	162	146	187	171	178	230	201	275	506	155	201	176
	PrecMin	868	28	20	19	30	34	49	62	38	11	0	12	26
	PrecStd	174	36	31	42	34	35	50	42	59	103	43	47	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.0	23.8	27.4	31.5	33.0	36.0	36.9	35.5	33.6	30.3	25.9	23.4
		MCWB	16.7	17.2	17.0	19.6	22.6	24.3	25.3	25.1	23.2	22.1	19.5	18.4
	2%	DB	19.1	21.0	24.4	28.6	31.0	33.7	34.7	33.8	31.8	27.8	22.8	20.6
		MCWB	15.4	15.0	15.7	19.0	21.7	23.9	25.2	24.5	23.2	21.0	17.2	17.3
	5%	DB	17.1	18.1	22.1	26.4	29.2	32.3	33.2	32.4	30.1	25.8	21.0	18.0
		MCWB	13.7	13.4	14.6	17.7	21.2	23.7	24.8	24.2	22.6	19.9	16.6	15.4
	10%	DB	14.1	15.9	19.4	24.0	27.6	30.8	32.0	31.1	28.3	23.7	18.6	15.8
		MCWB	10.6	11.3	13.7	17.1	20.1	23.1	24.2	23.7	22.0	18.8	15.2	13.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	18.8	18.8	21.8	24.1	26.1	27.1	26.9	25.5	24.5	21.0	20.0
		MCDB	20.5	22.1	23.9	27.3	30.0	32.6	33.1	32.1	30.4	27.1	23.2	22.3
	2%	WB	16.5	16.7	17.6	20.4	23.0	25.2	26.3	26.0	24.7	22.8	19.0	17.7
		MCDB	18.8	19.6	21.7	26.1	28.5	31.2	32.7	31.2	29.0	26.3	21.9	19.7
	5%	WB	14.1	14.6	16.4	19.2	22.2	24.6	25.7	25.4	24.1	21.4	17.4	15.9
		MCDB	16.4	17.9	21.3	24.4	27.2	30.2	31.5	30.6	27.9	24.4	19.4	18.0
	10%	WB	11.3	11.5	14.7	18.0	21.4	24.0	25.1	24.8	23.3	19.8	15.6	13.3
		MCDB	13.7	14.9	18.6	22.5	25.9	29.1	30.4	29.3	26.5	22.7	18.5	15.5
Mean Daily Temperature Range	5% DB	MDBR	10.8	11.7	12.4	12.9	11.7	11.1	10.7	10.5	10.7	12.0	12.2	10.9
		MCDBR	13.3	14.8	15.0	14.6	13.2	12.5	12.0	11.8	12.4	13.0	13.9	12.1
		MCWBR	10.0	10.1	8.2	6.6	5.9	5.0	4.5	4.4	5.3	6.8	8.7	9.5
	5% WB	MCDBR	12.2	13.8	12.8	12.2	11.1	10.9	10.6	10.0	9.5	10.3	10.8	11.3
		MCWBR	10.4	10.8	8.9	6.6	5.5	4.9	4.4	4.3	4.6	6.0	8.2	9.8
Clear-Sky Solar Irradiance	taub	0.302	0.316	0.345	0.390	0.432	0.488	0.513	0.509	0.405	0.354	0.321	0.306	
	taud	2.531	2.489	2.437	2.323	2.233	2.127	2.075	2.072	2.355	2.491	2.537	2.556	
	Ebn at Noon	895	922	920	891	854	805	780	773	846	866	865	867	
	Edh at Noon	78	92	106	126	140	155	163	160	114	91	77	72	
All-Sky Solar Radiation	RadAvg	2.38	3.12	4.12	5.23	5.89	6.35	6.03	5.45	4.44	3.52	2.66	2.10	
	RadStd	0.19	0.35	0.33	0.40	0.51	0.42	0.30	0.29	0.40	0.42	0.21	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	N/A
(47)	Regional (67 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	+107

Nomenclature: See separate page